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SEMICONDUCTOR TECHNICAL DATA

TXR316 Series

4 Amp RMS Sensitive Gate Triacs

The TXR316 series are sensitive gate triacs designed for general-purpose AC power switching applications. These devices feature low gate trigger current, high commutating dv/dt capability, and are suitable for use in motor control, lighting dimmers, static relays, and temperature control systems. Available in three sensitivity grades (TXR316, TXR316A, TXR316B) to match a wide range of gate drive requirements.

TO-220AB (TO-220-3 Full Pack)	Pin	Symbol	Description
	1	MT2	Main Terminal 2 (Anode)
	2	Gate	Gate Terminal
	3	MT1	Main Terminal 1 (Cathode / Reference)

MAXIMUM RATINGS

Rating	Symbol	TXR316	TXR316A	TXR316B	Unit
Repetitive Peak Off-State Voltage	V_{DRM}	400	600	800	V
RMS On-State Current ($T_c = 85^\circ\text{C}$)	$I_{\text{T(RMS)}}$	4.0	4.0	4.0	A
Peak Non-Repetitive Surge Current (1 cycle, 50/60 Hz)	I_{TSM}	30	30	30	A
Peak Gate Current	I_{GM}	0.5	0.5	0.5	A
Peak Gate Voltage	V_{GM}	5	5	5	V
Peak Gate Power	P_{GM}	5	5	5	W
Average Gate Power	$P_{\text{G(AV)}}$	0.5	0.5	0.5	W
Operating Junction Temperature Range	T_j	-40 to +125	-40 to +125	-40 to +125	$^\circ\text{C}$
Storage Temperature Range	T_{stg}	-40 to +150	-40 to +150	-40 to +150	$^\circ\text{C}$

THERMAL CHARACTERISTICS

Characteristic	Symbol	Max	Unit
Thermal Resistance, Junction to Case	$R_{\text{th(j-c)}}$	3.0	$^\circ\text{C/W}$
Thermal Resistance, Junction to Ambient	$R_{\text{th(j-a)}}$	100	$^\circ\text{C/W}$

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Characteristic	Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS					
Repetitive Peak Off-State Current ($V_{\text{DM}} = V_{\text{DRM}}$, $T_j = 125^\circ\text{C}$)	I_{DRM}		0.02	0.5	mA
TXR316 ($V_{\text{DM}} = 400\text{ V}$)			0.02	0.5	
TXR316A ($V_{\text{DM}} = 600\text{ V}$)			0.02	0.5	
TXR316B ($V_{\text{DM}} = 800\text{ V}$)			0.05	0.8	

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted) (Continued)

Characteristic	Symbol	Min	Typ	Max	Unit
ON CHARACTERISTICS					
On-State Voltage ($I_{TM} = 4\text{ A}$, $T_j = 25^\circ\text{C}$)	V_{TM}		1.55	1.80	V
On-State Voltage ($I_{TM} = 4\text{ A}$, $T_j = 125^\circ\text{C}$)	V_{TM}		1.45	1.70	V
Holding Current ($V_D = 12\text{ V}$, $I_G = 0$)	I_H		5	15	mA
Latching Current ($V_D = 12\text{ V}$, $I_G = 0$)	I_L		8	20	mA

Characteristic	Symbol	TXR316	TXR316A	TXR316B	Unit
GATE CHARACTERISTICS					
Gate Trigger Current ($V_D = 12\text{ V}$, $R_L = 100\text{ Ohm}$)	I_{GT}				
Quadrant I (III)		2	5	10	mA
Gate Trigger Voltage ($V_D = 12\text{ V}$, $R_L = 100\text{ Ohm}$)	V_{GT}				
Quadrant I (III)		0.7	0.7	0.7	V
Gate Non-Trigger Voltage ($V_D = V_{DRM}$)	V_{GD}	0.2	0.2	0.2	V

Characteristic	Symbol	Min	Typ	Max	Unit
SWITCHING CHARACTERISTICS					
Critical Rate of Rise of Off-State Voltage	dv/dt	5.0	10		V/us
Commutating dv/dt ($di/dt = 4.5\text{ A/ms}$, $T_j = 125^\circ\text{C}$)	$(dv/dt)_c$	1.5	4.5		V/us
Turn-On Time ($I_{TM} = 4\text{ A}$, $I_G = 50\text{ mA}$)	t_{gt}		2.0	5.0	us
Turn-Off Time (at rated V_{DRM} and I_{TM})	t_q		50		us

Note 1: All electrical characteristics are measured at $T_A = 25^\circ\text{C}$ unless otherwise specified. Voltage and current values are referenced to MT1 (Pin 3). Quadrant definitions follow standard IEC 60747 convention: Quadrant I (MT2 positive, Gate positive) and Quadrant III (MT2 negative, Gate positive).

Note 2: I_{GT} values apply to both Quadrant I and Quadrant III gate triggering. For Quadrant IV operation (MT2 negative, Gate negative), the gate trigger current may be higher. Design should ensure adequate gate drive for all operating quadrants.

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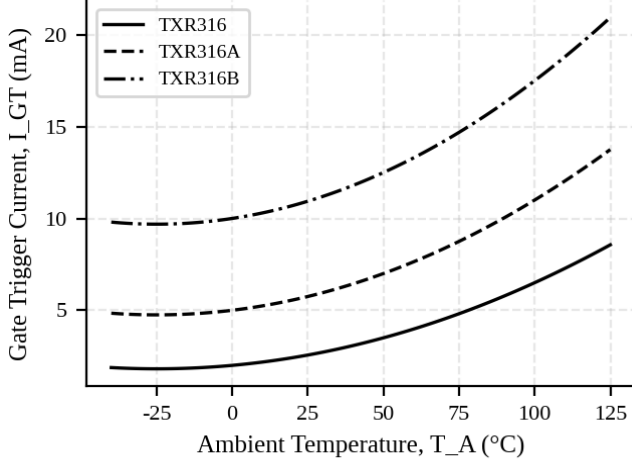
Figure 1. Gate Trigger Current vs Temperature

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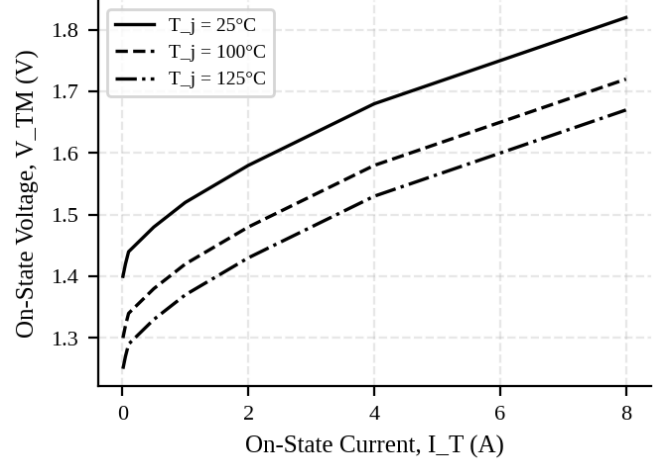
Figure 2. On-State Characteristics

Figure 2. On-State Voltage Characteristics

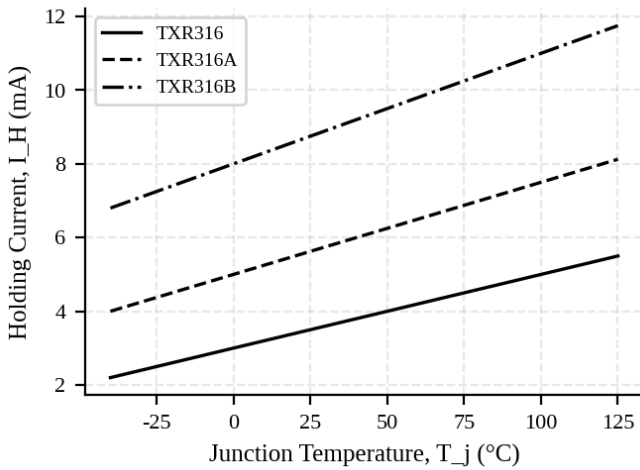
Figure 3. Holding Current vs Temperature

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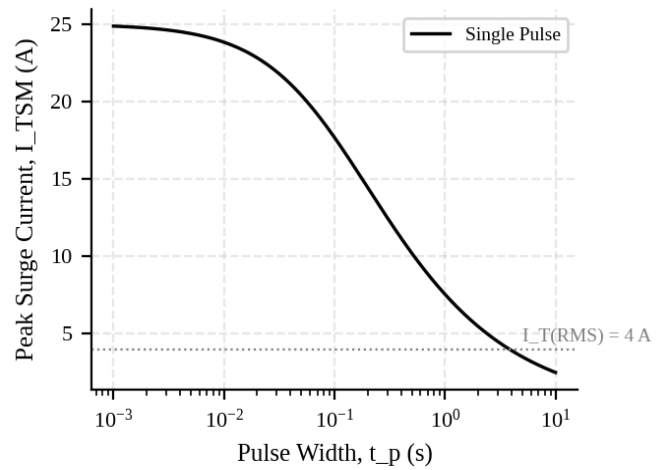
Figure 4. Surge On-State Current

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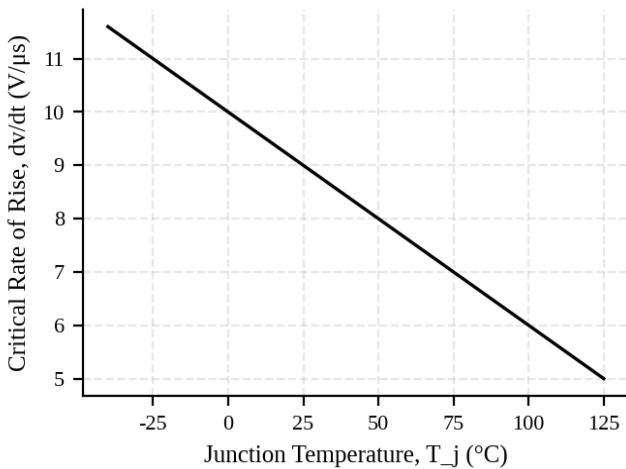
Figure 5. Critical Rate of Rise of Off-State Voltage

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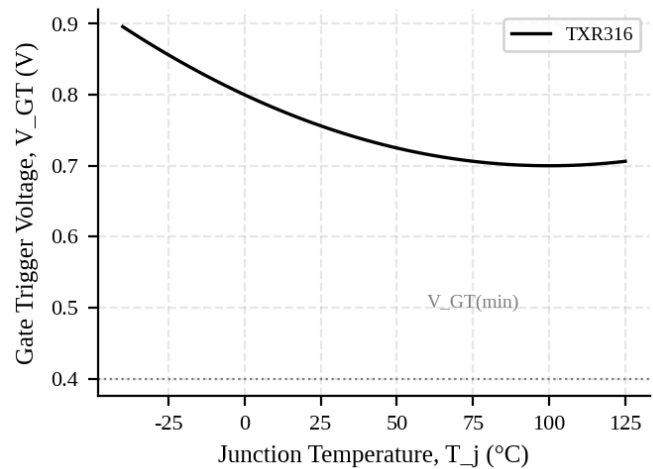
Figure 6. Gate Trigger Voltage vs Temperature

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Figure 7. Gate Power Derating

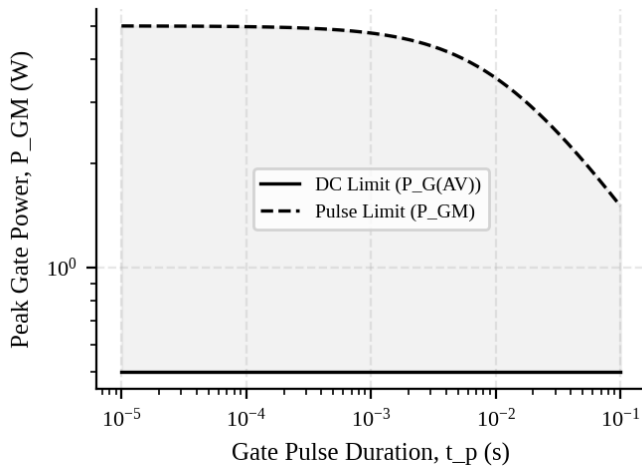


Figure 7. Gate Power Derating

Figure 8. Thermal Response

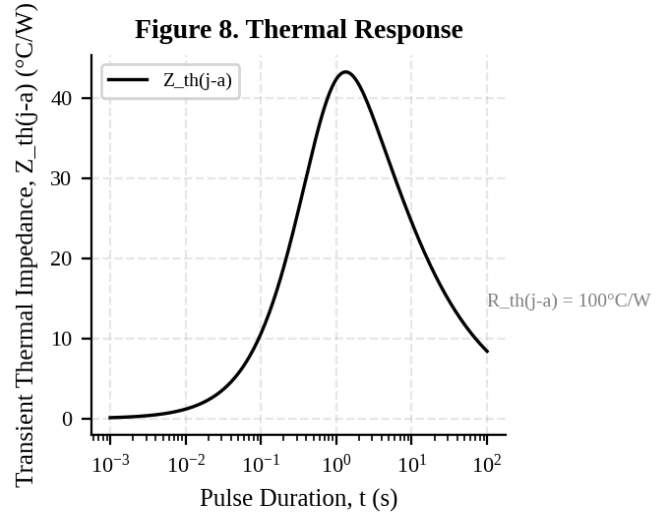


Figure 8. Thermal Response

Figure 9. Commutating dv/dt

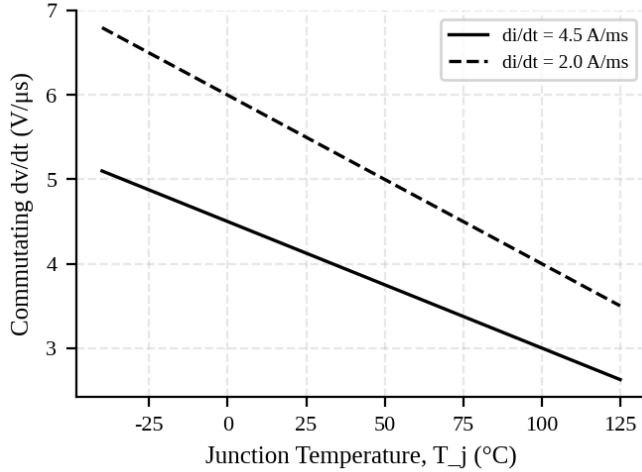


Figure 9. Commutating dv/dt

Thermal Limit

Figure 10. Safe Operating Area

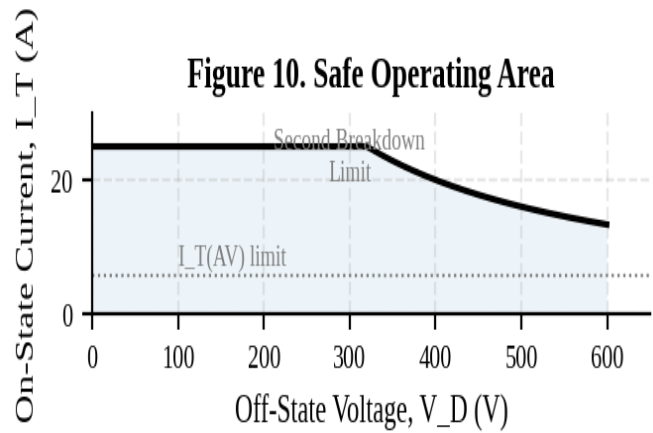


Figure 10. Safe Operating Area

Safe Operating Area Note: The safe operating area curves indicate I_T - V_D limits of the triac that must be observed for reliable operation. Load lines for specific circuits must fall below the limits indicated by the applicable curve. The data is based upon $T_{j(pk)} = 125^\circ\text{C}$. Pulse curves are valid for duty cycles to 10% provided $T_{j(pk)}$ does not exceed 125°C . At high case or ambient temperatures, thermal limitations will reduce the power that can be handled to values less than the limitations imposed by second breakdown.

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PACKAGE DIMENSIONS

NOTES:

1. Dimensioning and tolerancing per ANSI Y14.5M, 1982.
2. Controlling dimension: INCH.
3. Package outline conforms to JEDEC TO-220AB specification.
4. Leads are plated and solderable per MIL-STD-202 Method 208.

Dim	Min (in)	Max (in)	Min (mm)	Max (mm)	TO-220AB (Style 221A)
A	0.610	0.650	15.50	16.50	
B	0.390	0.430	9.90	10.90	Pin 1: MT2 Pin 2: Gate Pin 3: MT1
C	0.140	0.180	3.56	4.57	
D	0.025	0.039	0.64	0.99	
E	0.015	0.021	0.38	0.53	
F	0.095	0.105	2.41	2.67	
G	0.300		7.62		
H	0.100	0.120	2.54	3.05	
J	0.115	0.145	2.92	3.68	
K	0.490	0.530	12.45	13.46	
L	0.040	0.060	1.02	1.52	
N	0.500		12.70		
Q	0.100	0.120	2.54	3.05	

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